

UVERA

Trust you can verify.

A Calibrated, Explainable and Human-in-the-Loop AI Trust Layer for Mobile Financial Services in Bangladesh

Pausing scams before the money moves, planning agent cash, and turning alerts into evidence-backed cases

Competition	AI DEV FEST 2026 — AI Hackathon, DIU CPC × upay (three-day development period)
Track	07 · Open Innovation
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Source code	https://github.com/ABRUBAB/takaresolve-ai-hackathon-2026 (MIT licence)
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Abstract. Mobile financial services (MFS) have brought millions of people in Bangladesh into digital finance, but they also opened a fast channel for social-engineering scams: the customer is persuaded to *authorise* the transfer, and the money hops through mule wallets before it is cashed out at an agent or through a merchant QR code. We present **UVERA**, a trust layer that works beside an MFS and helps the three people who can stop such a scam: the customer before pressing *Confirm*, the cash-out agent, and the operations analyst. UVERA combines seven AI components: a calibrated LightGBM *Pause Check* with a conformal “not sure” state and exact TreeSHAP reasons; a Bangla, Banglish and English scam-text classifier; LightGBM-quantile and Chronos-2 forecasters for customer cash flow and agent liquidity; a QR cash-out detector that fuses supervised and unsupervised scores; a time-respecting money-path case linker; and a Gemini brief bounded by a deterministic validator. A rule-based policy sits on top: it never blocks the customer and leaves every high-impact action to a person. All models are trained and evaluated in nine reproducible Kaggle notebooks on a seeded synthetic world of 2.2 million transactions with hidden scam processes, using time- and customer-disjoint splits and a single test pass. At the same 5% alert budget the Pause Check catches **86.2%** of scams versus 28.2% for a common rule; case linking removes **47.7%** of analyst work items while the top five cases hold 94% of real scam alerts; agent cash planning reaches **92.5%** of the best achievable gain; and **no** prompt-injection attack reached a user. A working website and API demonstrate the full flow.

Keywords: mobile financial services; scam detection; authorised push-payment fraud; probability calibration; conformal prediction; explainable AI; time-series forecasting; grounded language models; human-in-the-loop; Bangladesh.

UVERA at a glance (test data the models never saw)

86.2%

scams caught at a 5% alert rate
(simple rule: 28.2%)

0.986

scam-text PR-AUC on an
unseen
writing style (Bangla 0.996)

377 → 197

alerts joined into cases;
top 5 cases hold 94% of scams

0%

prompt-injection attacks
shown
to users (raw LLM: 10%)

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1. Introduction

Mobile money has changed how people save, pay and support each other. Large studies link it to lower transaction costs, better risk sharing between households, and long-run reductions in poverty [1, 2], and it is a main driver of account ownership in developing economies [3]. In Bangladesh, mobile wallets reach customers who never used a bank, through a dense network of cash-in and cash-out agents and, increasingly, through interoperable Bangla QR merchant payments.

The same speed that makes mobile money useful makes it attractive to scammers. The most damaging attacks today do not steal a password: they persuade the customer to send the money themselves. A “wrong-number” refund request, a fake customer-care call, a lottery that needs a fee, or a guaranteed-profit group all end with the customer pressing *Confirm*. Because the transfer is authorised, systems built to catch stolen credentials see a normal payment. The money then moves within hours through *mule* wallets and is cashed out at an agent or through a merchant QR code that acts as a hidden cash machine. On the other side of the network, agents run short of cash on busy days, and operations teams receive many separate alerts that belong to the same scam, while new regulation sets strict deadlines for handling Bangla QR payment disputes [4, 5].

UVERA (from the Latin *vera*, “true”) is our answer for Track 07: a trust layer that works beside a wallet and gives each of these people the right help at the right moment. Its guiding idea is simple: *pause before the money moves, show the evidence, and let a person decide*. This report makes the following contributions:

- C1. A pre-transfer Pause Check** (AI-1) that turns 19 past-only behaviour signals into a calibrated scam probability, says “not sure” through class-conditional conformal prediction and a novelty check, and explains itself with exact TreeSHAP reasons, while never blocking the customer.
- C2. A multilingual scam-text check** (AI-2) for Bangla, Banglish and English, tested on a writing style it never saw.
- C3. Calendar-aware forecasting** for customer cash flow (AI-3) and agent liquidity (AI-4), compared fairly with the Chronos-2 foundation model [6], including a simulated *perfect-knowledge limit* that shows how close the agent forecast is to the best achievable.
- C4. A QR cash-out shield** (AI-5) and a **money-path case linker** (AI-6) that turn hundreds of alerts into a small queue of evidence-backed cases with a regulatory deadline clock.
- C5. A grounded brief** (AI-7): Gemini [7] rewrites the evidence in Bangla and English, and a deterministic validator stops any new number, link, promise or injected instruction from reaching a user.
- C6. A reproducible evaluation** on a seeded synthetic world (nine Kaggle notebooks, fixed seeds, single test pass, SHA-256 artifact manifest) and a **working prototype** with customer, agent, operations and Trust Center areas.

Section 2 states the problem; Sections 3–6 present the idea, architecture, data and AI approach; Sections 7–8 give the protocol and results; Sections 9–12 describe the prototype, responsible-AI design, impact and path to deployment.

2. Problem Statement and Users

Problem. In an MFS, a scam is decided in the few seconds before a customer confirms a transfer, but the evidence that it is a scam is spread across the receiver’s recent history, the money’s path after the transfer, and the shops and agents where it is cashed out. No single person sees all of it, and today’s tools either stay silent or block without explaining why.

Following the guideline’s idea-development framework, Table 1 links each user to a concrete problem, today’s baseline, and the measurable change UVERA aims for.

Table 1. Users, problems and the measured change. Baselines are the methods a basic system would use today.

User	Problem	Today’s baseline	UVERA’s measured change
Customer	Authorised scam transfers; little help to judge a suspicious SMS or the week ahead	Rule “large amount to a new receiver”; no explanation	86.2% vs 28.2% of scams caught at the same alert rate; reasons in Bangla
Agent	Runs out of cash on busy days, or holds too much	Usual cash on hand	Days short of cash 21.0% → 12.6%
Operations analyst	Many alerts about the same scam; legal deadlines	One alert = one work item	47.7% fewer items; deadline clock
Ecosystem	Merchant QR used as a hidden cash-out to skip fees	Peer z-score	92.5% precision in the weekly top-20 review

Why now. Three developments make this practical for a student team in 72 hours: gradient-boosted trees with exact explanations are fast and strong on tabular data [8–11]; calibration and conformal prediction give honest probabilities and a principled “I don’t know” [12, 13]; and pretrained time-series and language models are available as zero-shot tools [7, 14], provided their output is checked. Fraud research also shows that realistic, time-ordered evaluation and recalibrated probabilities matter as much as the classifier itself [15, 16].

Design requirements. From the guideline’s responsible-AI principles we derived six requirements that every component must meet: **R1** never block a customer automatically; **R2** output calibrated probabilities and say “not sure” when evidence is mixed; **R3** show the main reasons in Bangla and English; **R4** keep business rules outside the models; **R5** beat a simpler baseline on data the model never saw; **R6** use only synthetic or open data.

3. Proposed Idea: the UVERA Trust Layer

UVERA is a trust layer, not a replacement for the wallet. It checks each event (a draft transfer, a message, a day of cash flow, a week of QR payments), estimates risk with an honest uncertainty, explains it in plain Bangla and English, and hands the decision to a person: the customer, the agent or the analyst.

One website has four doors: the **customer** area (Pause Check, Scam Text Check, Cash-Flow Guardian), the **agent** area (cash to hold, peer comparison, area QR risk), the **operations** area (case queue, case workspace, QR watchlist) and the **Trust Center**, where judges and auditors see every metric with its source. Six design principles turn R1–R6 into product behaviour: *never block* (the customer can always send anyway; a hold on a large high-risk transfer needs staff approval); *say “not sure”* (mixed or unusual cases go to a person); *show evidence* (reasons, model and data version, trace id); *Bangla first*; *rules stay rules* (limits and deadlines live in configuration files); and *label everything* as a model estimate, a business rule or generated wording.

Figure 1 shows the end-to-end workflow: models and thresholds are built and frozen offline; at run time each transfer is scored and routed to one of three outcomes; a learning loop is planned for a pilot with governed real data.

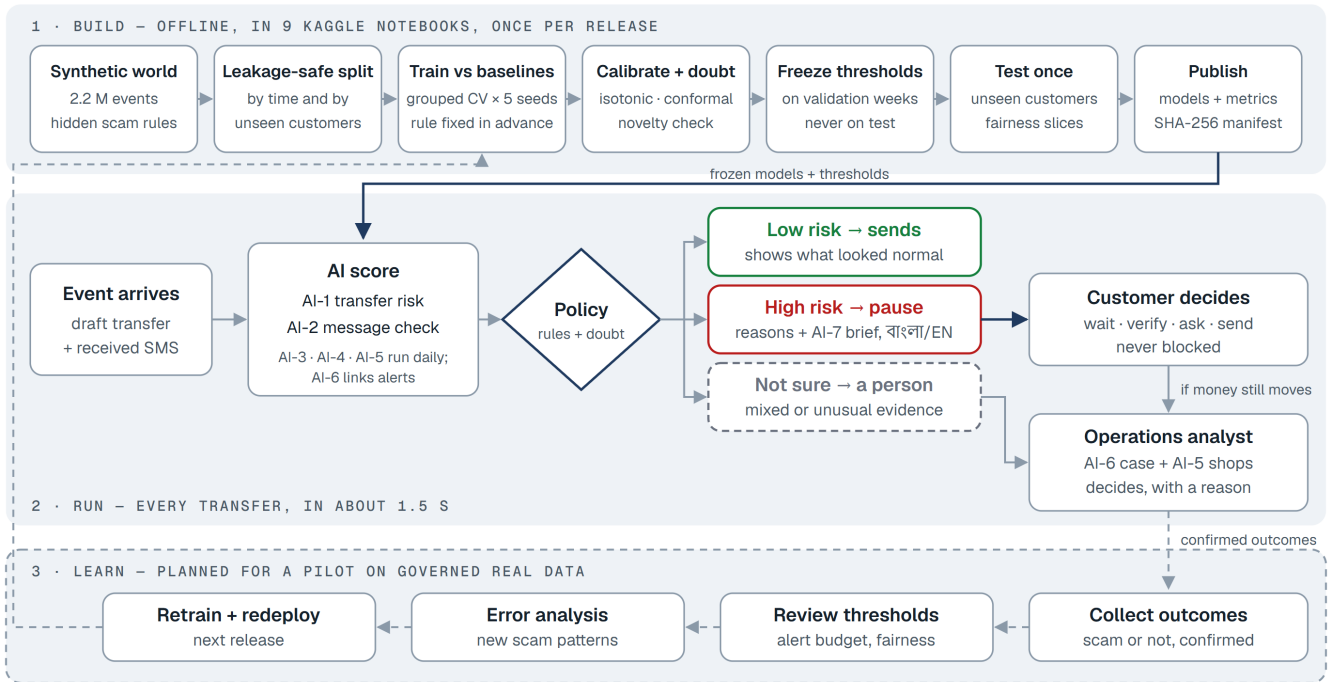


Figure 1. UVERA workflow. *Build*: nine Kaggle notebooks create the synthetic world, train against baselines, calibrate, freeze thresholds on validation weeks and test once. *Run*: the policy sends low-risk transfers, pauses high-risk ones with reasons, and routes “not sure” to a person; money that still moves reaches the operations analyst as a linked case. *Learn*: the planned feedback loop for a pilot (not part of the prototype).

4. System Architecture and Implementation

Figure 2 shows the architecture. Six input streams feed six scoring components; their scores, uncertainty and reasons go to a *decision policy* that is a configuration file rather than a model, so a threshold can change without retraining. AI-7 then rewrites the evidence for the person who decides. Table 2 lists exactly what was built and tested for each component.

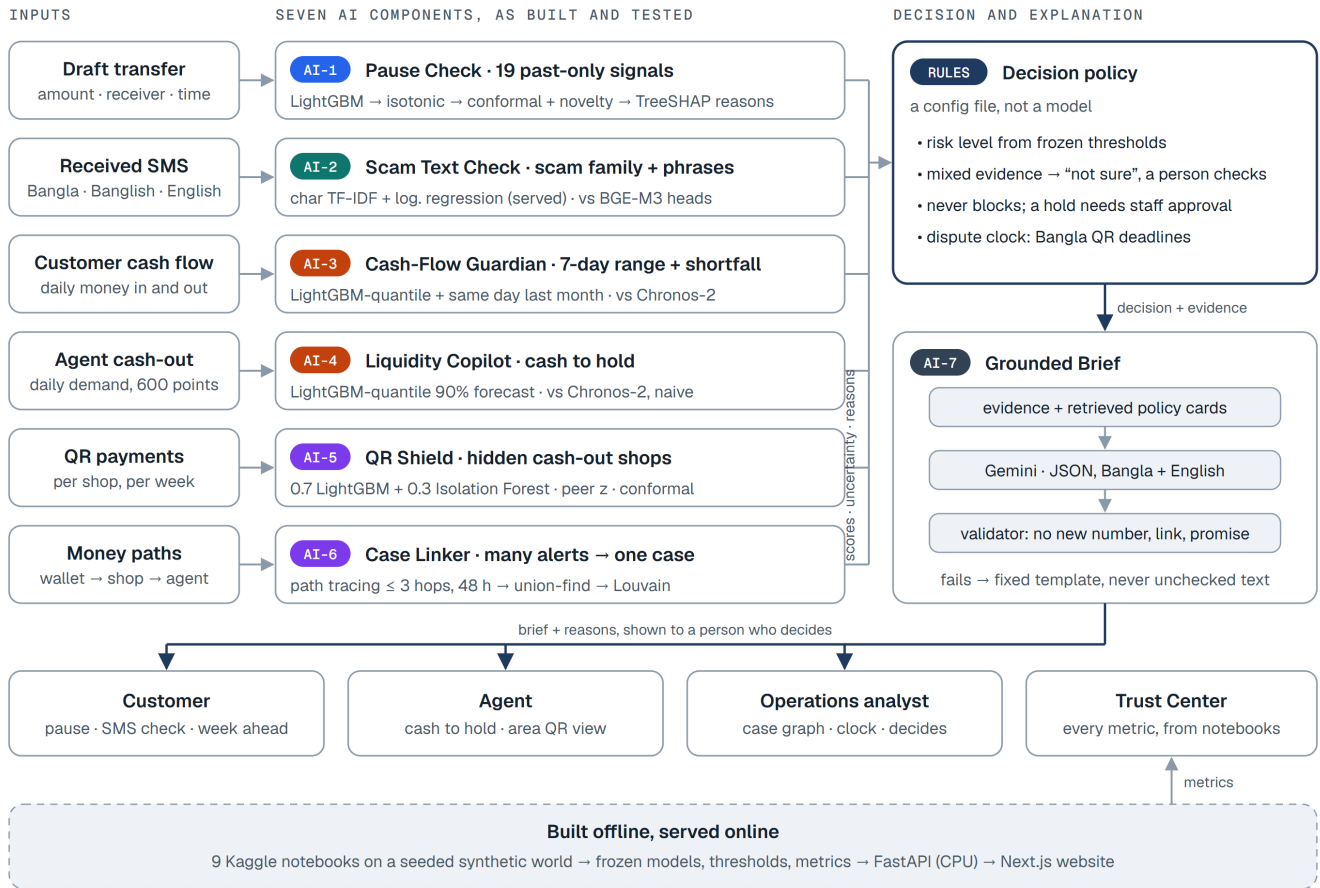


Figure 2. UVERA architecture as implemented. Each AI has its own colour; arrows carry scores, uncertainty and reasons to a rule-based policy, then to the grounded brief, and finally to a person. Everything is built offline in nine Kaggle notebooks and served by a CPU API to a Next.js website.

Table 2. The seven AI components as built and evaluated. “Served” is the model the live API uses.

Component	Task	Model (served in bold)	Uncertainty	Explanation
AI-1 Pause Check	Scam risk of a draft transfer	LightGBM ; vs XGBoost, CatBoost, logistic regression, rule	Isotonic; Mondrian conformal; novelty	Exact TreeSHAP, counterfactual amount
AI-2 Scam Text	Scam or not, scam family	char TF-IDF + logistic regression ; vs BGE-M3 + LR, BGE-M3 + evidential head	Isotonic verdict; “not sure” band	Phrase occlusion
AI-3 Cash-Flow	7-day net flow, shortfall risk	LightGBM-quantile with weekly and same-day-last-month lags; vs Chronos-2, seasonal-naive	q10/q50/q90; empirical shortfall probability	Forecast band, heavy weeks
AI-4 Liquidity	Agent cash-out demand	LightGBM-quantile ; vs Chronos-2, seasonal-naive	q10/q50/q90; perfect-knowledge limit	Cash above usual
AI-5 QR Shield	QR used as hidden cash-out	0.7 LightGBM + 0.3 Isolation Forest (rank fusion); vs peer z-score	Isotonic; conformal grey state	Top reasons, peer comparison
AI-6 Case Linker	Alerts → cases	Time-respecting path tracing + union-find , Louvain communities	Transparent chain score	Evidence graph E1–E5
AI-7 Grounded Brief	Bangla/English brief	Gemini (structured JSON) + validator , template fallback	Inherits upstream state	Cites evidence and policy-card ids

Software. All learning code lives in a tested Python package (`uvera_ml`); the nine notebooks only orchestrate it, pull the code from GitHub at a recorded commit, and package their outputs. A FastAPI service loads the exported artifacts (LightGBM text models, JSON calibrators and thresholds, Parquet score tables) and serves the website. If the API is offline, the website plays back responses recorded from the real API, so the demonstration never breaks.

The stack is Python (LightGBM, XGBoost, CatBoost, scikit-learn [17], PyTorch, sentence-transformers, Chronos-2, NetworkX) on Kaggle CPU and T4 GPU; the Gemini API with structured JSON output; FastAPI with JWT access control, rate limits and trace ids; and Next.js, TypeScript, Tailwind CSS and React Three Fiber on Vercel. Each run records its Git commit, package versions and a SHA-256 manifest of all 28 model files.

5. Data: a Seeded Synthetic MFS World

No real customer data was used or needed. NB00 generates a seeded world (seed 42) of **20,000** customers, **600** agents and **2,500** merchants over **120** days, producing **2,203,725** events: salaries, remittances, cash-in and cash-out, person-to-person transfers, QR payments, bills and top-ups, with weekly cycles, pay days, month-end pressure and a festival week. Every assumption is written in configuration files.

Hidden processes, not label rules. Scams come from a hidden process: susceptible victims (newer and USSD users are more susceptible, but susceptibility is never a feature) are approached by one of six scam families and pay one of 200 mule wallets, which forward the money within 48 hours to other mules, agents, or QR shops (Fig. 3). Four disguise families make merchant QR codes act as cash machines: round amounts (A), split bursts under limits (B), cash-in pass-through (C) and a rotating payer ring (D); family D is hidden from all training to test generalisation. Honest look-alikes (rent to a new receiver, festival family transfers, fixed-price shops) and **10% label noise** (373 scams unreported, 373 false reports) keep the task from being trivial.

Leakage guard and splits. No single feature may separate scams with AUC above 0.80 (the highest is 0.781). Data are split by time and by customer: days 14–69 train, 70–84 calibration, 85–99 validation, 100–119 test; the test customers are a 30% group never used in training. Every notebook regenerates the world and checks its hash, so all results refer to identical data.

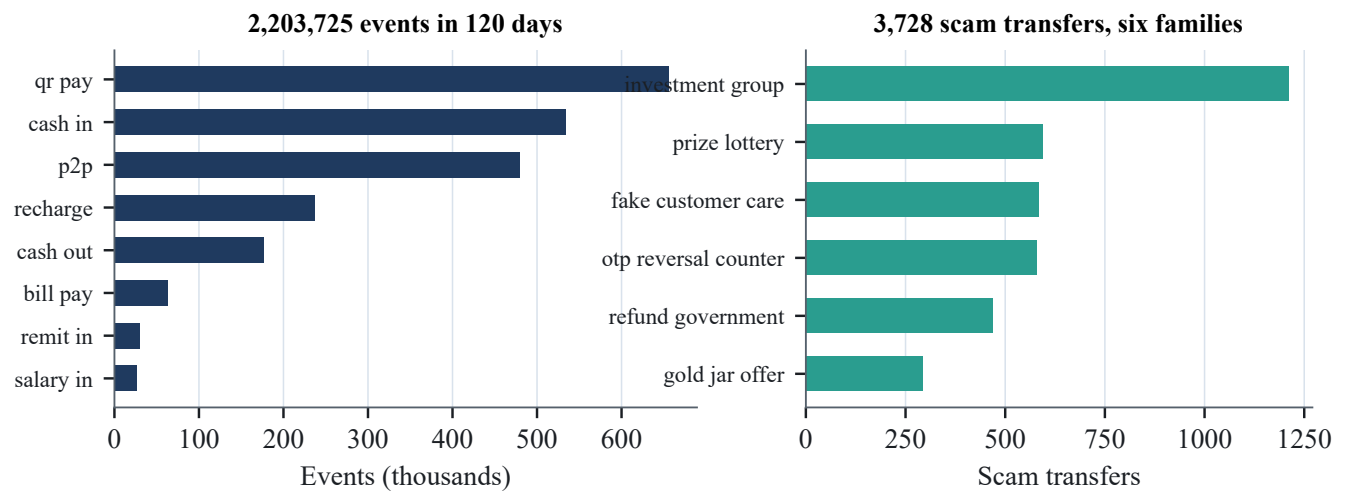


Figure 3. The synthetic world: events by type (left) and the 3,728 scam transfers by family (right).

Table 3. Synthetic world summary (from reports/data_card_stats.json).

Customers (urban / semi-urban / rural)	20,000	Scam transfers (share of P2P)	3,728 (0.78%)
Channel: app / USSD	14,199 / 5,801	Scam cases / mule wallets	3,100 / 200
Language: Bangla / Banglish / English	7,961 / 6,960 / 5,079	Disguised QR merchants (A/B/C/D)	46 / 35 / 34 / 26
Agents / merchants	600 / 2,500	P2P amount median / p90 / p99 (Tk)	670 / 4,358 / 13,000

6. AI Approach: Seven Components

6.1. AI-1 Pause Check

Features. For a draft transfer, 19 signals use only information available before it: nine about the sender (amount, amount relative to the last 30 days and to the balance, tenure, transfers sent in 7 days, minutes since the last cash-in,

recent device change, recent PIN reset, first-time pair), six about the receiver (wallet age, number of distinct senders in 7 days, incoming count, out/in money ratio, cash-outs, transfers already received today) and four about the moment (hour, night, weekday, USSD).

Model selection. LightGBM, XGBoost, CatBoost [18], logistic regression and the rule are fitted on identical StratifiedGroupKFold splits (5 folds grouped by sender × 5 seeds, so no customer leaks across folds [19]) and compared by PR-AUC, which is more informative than ROC-AUC for rare positives [20]. A rule fixed in advance keeps LightGBM unless another model is better by a paired margin of at least 0.01 PR-AUC; the test set is never used for this choice.

Calibration and doubt. With raw score $f(x)$, the calibrated probability is

$$p(x) = g(f(x)), \quad g = \arg \min_{g \text{ non-decreasing}} \sum_{i \in \mathcal{D}_{\text{cal}}} (y_i - g(f(x_i)))^2, \quad (1)$$

fitted on the calibration window [12, 21]. A class-conditional split-conformal predictor [13, 22] is then fitted on the *separate* validation window: for each class c , $\hat{q}_c$ is the $\lceil (n_c + 1)(1 - \alpha) \rceil / n_c$ empirical quantile of the non-conformity scores $1 - p_c(x_i)$ of that class, and

$$C(x) = \{c \in \{0, 1\} : 1 - p_c(x) \leq \hat{q}_c\}, \quad \alpha = 0.10. \quad (2)$$

A transfer is shown as “not sure” when $C(x)$ contains both labels while its risk is not low, or when a robust novelty score (mean of the three largest robust $|z|$ values) exceeds its training 99.5th percentile; abstaining in this way is a form of selective classification [23].

Thresholds and reasons. Thresholds on the continuous score are frozen on validation: *amber* flags 5% and *red* (pause) 1% of normal validation transfers. Exact TreeSHAP contributions [11, 24] give the top three risk-raising reasons, or, for a low-risk draft, the facts that lowered risk; a counterfactual tells whether a smaller amount would change the outcome.

6.2. AI-2 Scam Text Check

A corpus of **11,092** synthetic messages (8,636 written by Gemini, 2,456 from templates; Bangla 3,732, Banglish 3,670, English 3,690) covers six scam families and six legitimate types. Two *writing-style families* are kept apart: style A for training and cross-validation, style B (chat-like, code-mixed, typos, emoji) only for testing. Three models are compared with StratifiedGroupKFold grouped by template and generation batch: character TF-IDF (2–5-grams) with logistic regression, BGE-M3 embeddings [25] with logistic regression, and BGE-M3 with an evidential head [26]. The live API serves the TF-IDF model because it runs on a CPU-only server; its verdict probability is isotonic-calibrated and phrases are highlighted by occlusion (remove a phrase, measure the drop). Messages that contain instructions aimed at an AI are flagged.

6.3. AI-3 Cash-Flow Guardian and AI-4 Agent Liquidity

Both forecast seven days ahead with three methods compared on rolling origins [27]: a seasonal-naïve baseline, a global LightGBM quantile model with lags of 7–28 days (so no recursion is needed), and the zero-shot Chronos-2 foundation model with calendar covariates [6]. For customers (AI-3), salary, bills and remittances repeat on the same calendar day each month, so the LightGBM model also receives the value on the same day one and two months earlier. The winner is the lowest validation pinball loss

$$L_\tau(y, q) = \max(\tau(y - q), (\tau - 1)(y - q)), \quad \tau \in \{0.1, 0.5, 0.9\}, \quad (3)$$

[28, 29], with MASE [30] as tie-breaker. The probability that a customer’s balance ends the week below a Tk 200 floor is not obtained by adding daily quantiles. Instead, with 7-day sums μ (median) and width $s = q_{0.9} - q_{0.1}$, current balance B_t and floor F ,

$$\Pr(B_{t+7} < F) = \Pr(r \leq z), \quad r = \frac{S - \mu}{s}, \quad z = \frac{F - B_t - \mu}{s}, \quad (4)$$

where the distribution of r is the empirical distribution of normalised residuals on validation origins. The warning cut-off maximises F1 on validation within a 30% alert budget. AI-4 recommends holding the 90% forecast as cash.

To show how much room is left, we simulate a forecaster with *perfect same-day knowledge* of every customer balance and every mule cash-out, replaying each day’s random cash-outs 300 times; its loss L_{floor} is a lower bound for any real forecaster, and

$$G = \frac{L_{\text{naive}} - L_{\text{UVERA}}}{L_{\text{naive}} - L_{\text{floor}}} \quad (5)$$

is the share of the achievable improvement that UVERA reaches.

6.4. AI-5 QR Shield

Each merchant-week is described by 16 features computed only from issuer-visible data: round-amount share, bursts from the same payer, payments minutes after a cash-in, wallet drain, one-time and repeat payers, cross-merchant payers, night share, ticket size relative to the category, and robust peer z-scores against shops of the same *type, area and size*, so large honest shops are not over-flagged. A LightGBM model trained on families A–C is fused with an Isolation Forest [31] by rank, $s = 0.7 r_{\text{sup}} + 0.3 r_{\text{iso}}$, keeping some weight on the unsupervised detector so that new disguise styles can still surface. Scores are calibrated, a conformal grey state asks a person to check, and analysts review the top 20 merchant-weeks each week.

6.5. AI-6 Case Linker

From every high-risk or uncertain alert, AI-6 follows the money forward in time through transfers, QR payments and cash-outs, up to three hops within 48 hours. Alerts that share a downstream wallet are merged with union-find. Each case receives a transparent chain score

$$\text{score} = 1 - (1 - p_{\text{max}}) (1 - 0.15 \min(v - 1, 3)) (1 - 0.5 \phi) (1 - 0.3 q), \quad (6)$$

where p_{max} is the highest AI-1 probability (capped at 0.99), v the number of victims, ϕ the share of money moved on, and q whether a QR Shield-flagged shop is reached. Louvain communities [32] reveal sub-rings. The case workspace adds a *Dispute Clock*: deadlines from Bangladesh Bank’s 2026 Bangla QR dispute guideline as reported in the press [4, 5] (for example, two working days to accept or reject a complaint), stored as business rules, not model output.

6.6. AI-7 Grounded Brief

The decision and its reasons are packed into an *evidence object* together with policy cards retrieved by TF-IDF similarity, a retrieval-augmented design [33]. Gemini returns structured JSON (Bangla text, English text, cited evidence ids, cited card ids). A deterministic validator rejects any output that contains a number not present in the evidence, a link or phone number, a certainty phrase (“guaranteed”, “100% safe”), wording that contradicts a high risk, or that hides a “not sure” state; rejected output is replaced by a template built directly from the evidence. Because language models can be steered by instructions hidden in their input [34–36], the user-written note is marked as untrusted data in the prompt. If a Gemini model runs out of quota, it is paused and the next model in a fallback chain is used, so a run never stalls.

7. Experimental Protocol

All results come from the official Kaggle runs at a recorded commit; every shown number is read from the result files (reports/metrics_ai*.json). Table 4 summarises the protocol per component.

Table 4. Evaluation protocol. The test split is used once, after every choice is frozen.

AI	Model choice	Test	Main metrics
AI-1	Grouped CV, 5 folds × 5 seeds, paired rule	24,827 transfers of unseen customers, days 100–119 (181 scams)	PR-AUC, recall at 5% alerts, ECE, coverage
AI-2	Grouped CV on style A, 5 folds × 5 seeds	5,486 messages in the unseen style B; UCI SMS (external)	Verdict PR-AUC, macro F1
AI-3	Validation pinball loss	10,000 customers, two 7-day test origins	MASE, pinball, coverage, shortfall PR-AUC
AI-4	Validation pinball loss	600 agents, two 7-day test origins	MASE, days short of cash, limit share G
AI-5	Grouped CV by merchant, 5 × 5	2,559 merchant-weeks of unseen merchants; family D unseen	Weekly precision@20, recall by family
AI-6	Fixed design	Alerts from AI-1 on the test window	Items removed, top- k capture, linking
AI-7	Fixed validator	140 briefs + 30 prompt-injection attacks	Pass rate, injection shown

8. Results

Table 5 summarises the headline results; the subsections give the detail. All data are synthetic: the results show that the methods work end to end, not real-world accuracy.

Table 5. Headline results on test data the models never saw (from `reports/summary.md`).

AI	What is measured	UVERA	Baseline
AI-1 Pause Check	Scams caught when 5% of transfers are flagged	86.2%	28.2%
AI-1 Pause Check	PR-AUC on customers never seen in training	0.651	0.048
AI-1 Pause Check	False pauses per 1,000 normal transfers	8.9	13.5
AI-1 Pause Check	Calibration error (ECE)	0.002	—
AI-2 Scam Text	Verdict PR-AUC, unseen writing style (served model)	0.986	—
AI-2 Scam Text	Scam-family macro F1, unseen writing style	0.855	—
AI-3 Cash-Flow	Forecast error, MASE (LightGBM-quantile)	0.383	0.716
AI-3 Cash-Flow	Shortfall warning PR-AUC (vs history)	0.500	0.390
AI-4 Liquidity	Forecast error, MASE (LightGBM-quantile)	0.604	0.931
AI-4 Liquidity	Days short of cash when holding the 90% forecast	12.6%	21.0%
AI-4 Liquidity	Share of the achievable gain over naive, G	92.5%	—
AI-5 QR Shield	Real cash-out shops among the 20 reviewed each week	92.5%	—
AI-5 QR Shield	Recall on a disguise family never seen in training	42.1%	—
AI-6 Case Linker	Fewer items for analysts (alerts → cases)	47.7%	—
AI-6 Case Linker	Real scam alerts inside the top 5 cases	94.0%	—
AI-7 Brief	Prompt-injection attacks shown to users	0.0%	10.0% raw

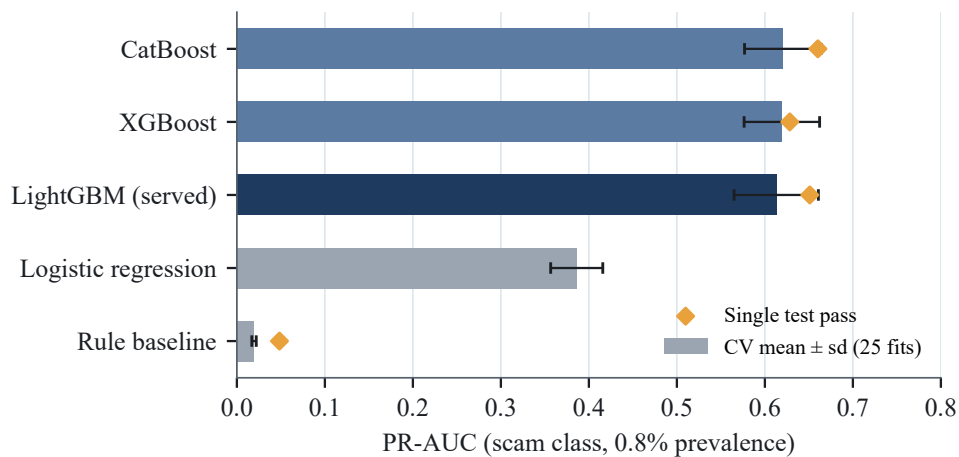
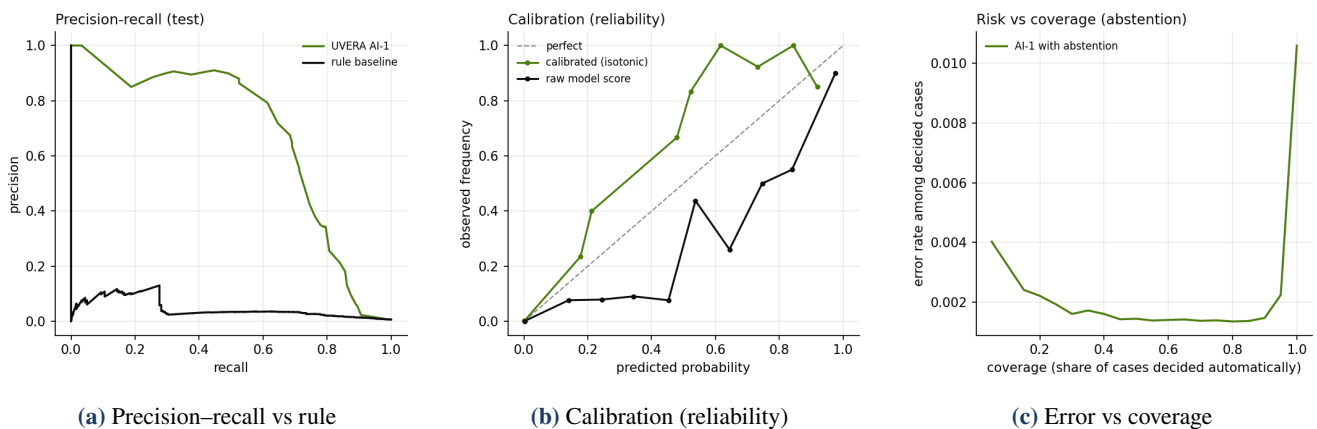
8.1. AI-1 Pause Check

On 24,827 transfers from customers never seen in training (181 scams, 0.73%), the Pause Check reaches PR-AUC **0.651** (95% bootstrap CI 0.567–0.726) against 0.048 for the rule, and ROC-AUC 0.931 against 0.831 (Table 6). At the same 5% alert budget it catches **86.2%** of scams (rule 28.2%) and Tk 692,660 of the Tk 744,224 scam money (rule Tk 437,300). At the pause threshold it raises 8.9 false pauses per 1,000 normal transfers, fewer than the rule’s 13.5, while catching 75.7% of scams. Isotonic calibration lowers the ECE from 0.0039 to 0.0017 (Fig. 5b). The conformal sets cover 94.2% of all transfers and 90.6% of scams (target 90%); only **0.6%** of transfers are shown as “not sure”, and their error rate is 6.3% against 1.0% for confident answers, so the abstention goes where the errors are (Fig. 5c). On customers seen in training the PR-AUC is 0.650, showing no gap between seen and unseen customers.

Table 6. AI-1 on unseen customers (test window), against the rule “Tk 5,000 or more to a new receiver” at the same alert rate.

Metric	UVERA AI-1	Rule baseline
PR-AUC (95% CI)	0.651 (0.567–0.726)	0.048
ROC-AUC	0.931	0.831
Recall / precision at a 5% alert rate	86.2% / 12.6%	28.2% / 4.1%
Scam money caught at a 5% alert rate	Tk 692,660	Tk 437,300
False alerts per 1,000 normal transfers (pause threshold)	8.9	13.5
Recall / precision at the pause threshold	75.7% / 38.5%	—
ECE calibrated (uncalibrated) / Brier score	0.0017 (0.0039) / 0.0036	—
Conformal coverage: all / scams / normal (target 90%)	94.2% / 90.6% / 94.2%	—
Shown as “not sure”; error when not sure vs confident	0.6%; 6.3% vs 1.0%	—

Model choice. In cross-validation, CatBoost (0.621) and XGBoost (0.619) were slightly ahead of LightGBM (0.613); the paired difference, CatBoost – LightGBM = +0.0076 (95% CI 0.0020–0.0132), was below the pre-registered margin of 0.01, so LightGBM was kept for exact TreeSHAP reasons and faster serving (Fig. 4). All three trees are far ahead of logistic regression (0.386) and the rule (0.020). A validation ablation shows that the receiver’s recent history carries most of the signal (PR-AUC 0.500 alone, 0.269 for the sender alone, 0.661 for all 19 features), which matches how mule wallets behave.

**Figure 4.** AI-1 model comparison: grouped cross-validation PR-AUC (mean ± sd over 25 fits) and the single test pass.**Figure 5.** AI-1 on unseen customers, produced by notebook NB01.

8.2. AI-2 Scam Text Check

On the unseen writing style, the served TF-IDF model reaches a verdict PR-AUC of **0.986** and scam-family macro F1 of 0.855 (Table 7); its uncertainty identifies wrong answers with AUROC 0.902. BGE-M3 with logistic regression was best in cross-validation (0.998) and has a higher macro F1 on style B (0.871), but needs a GPU-class model at serving time. By language (Fig. 8, right), Bangla is strongest (0.996) and Banglish weakest (0.979, F1 0.878). On the external UCI SMS Spam Collection [37, 38], a different task (English advertising spam), the served model reaches only 0.224 PR-AUC against a 13.4% base rate, while BGE-M3 reaches 0.538; we report this honestly as a limit of synthetic training text.

Table 7. AI-2 on the held-out writing style B (5,486 messages) and the external UCI set.

Model	CV PR-AUC (A)	Test PR-AUC (B)	Macro F1 (B)	UCI PR-AUC
Char TF-IDF + logistic regression (served)	0.992	0.986	0.855	0.224
BGE-M3 + logistic regression	0.998	0.977	0.871	0.538
BGE-M3 + evidential head	0.997	0.978	0.853	0.517

8.3. AI-3 Cash-Flow Guardian and AI-4 Agent Liquidity

Figure 6 compares the three forecasters. For customers, the same-day-last-month lag makes LightGBM-quantile the clear validation winner (MASE 0.752 vs 0.855 for Chronos-2 and 1.423 for seasonal-naive; validation weeks contain pay days), and on test weeks it keeps MASE **0.383** against 0.716 for the baseline, with its 80% range holding the truth 79.6% of the time. The shortfall warning reaches PR-AUC 0.500 against 0.390 for a customer’s own history and a lower Brier score (0.143 vs 0.163); at the validation cut-off it warns 32% of customer-weeks and catches 66% of shortfalls.

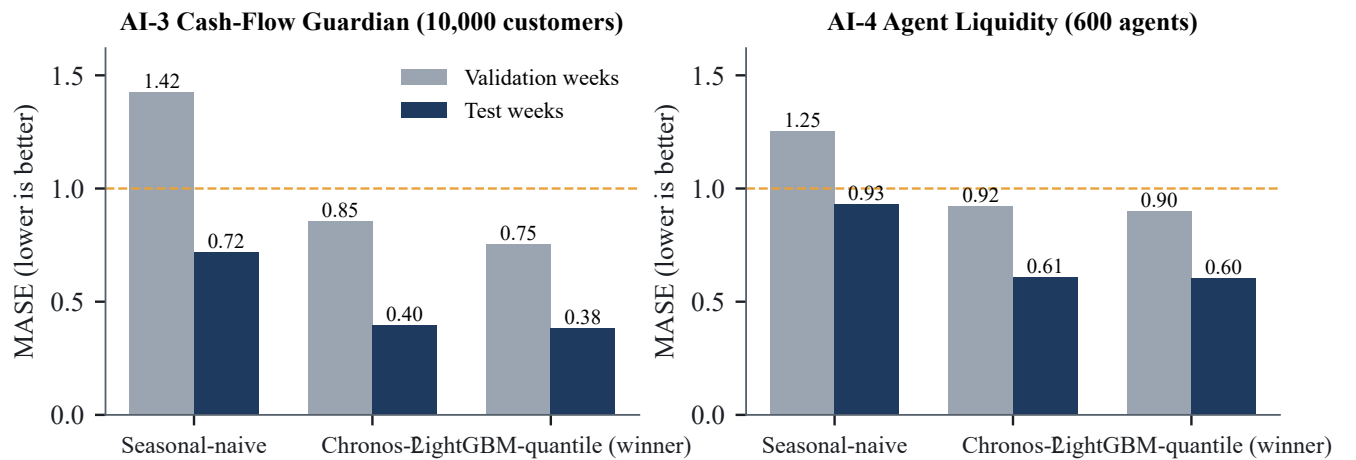


Figure 6. Rolling-origin backtests (MASE; the dashed line marks the naive error scale). The validation winner is marked.

For agents, LightGBM-quantile wins on validation and reaches test MASE **0.604** against 0.931. Holding the 90% forecast cuts days without enough cash from **21.0% to 12.6%** for Tk 4,920 more cash per day on average. Is more modelling worth it? The perfect-knowledge forecaster reaches a test pinball loss of Tk 2,162 against UVERA’s Tk 2,281 and the baseline’s Tk 3,732 (Fig. 7): UVERA already achieves $G = 92.5\%$ of the achievable improvement, and three stronger LightGBM designs (per-agent scaling, direct multi-horizon models, static agent features) changed the error by less than 1%. The remaining error is chance. For the same reason, which single day of an agent’s week runs short could not be predicted better than chance (precision 22.8% vs 21.0%), so the product does not flag individual days; ranking agents most at risk each week does beat history (precision 27.4% vs 20.4%).

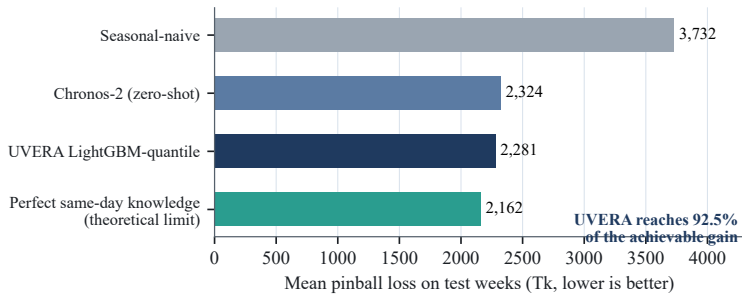


Figure 7. AI-4 against its theoretical limit. A simulated forecaster that knows every balance and every mule cash-out on the day cannot beat Tk 2,162; UVERA’s Tk 2,281 closes 92.5% of the gap from the naïve baseline.

8.4. AI-5 QR Shield

On 2,559 merchant-weeks of unseen merchants (179 disguised), **92.5%** of the 20 merchant-weeks reviewed each week are real cash-out shops; PR-AUC is 0.787 on seen disguise families against 0.314 for the peer z-score baseline. Recall is 84–91% for families A–C, and **42.1%** for the never-seen rotating ring D (Fig. 8, left). On family D the fused score (PR-AUC 0.122) beats both the supervised model (0.087) and the Isolation Forest (0.051), which justifies the fusion even though the supervised model alone is better on seen families (CV 0.806 vs 0.744). Only 5.9% of honest round-price shops are flagged; the false-positive rate is 11.0%, 9.2% and 7.9% for small, medium and large honest shops.

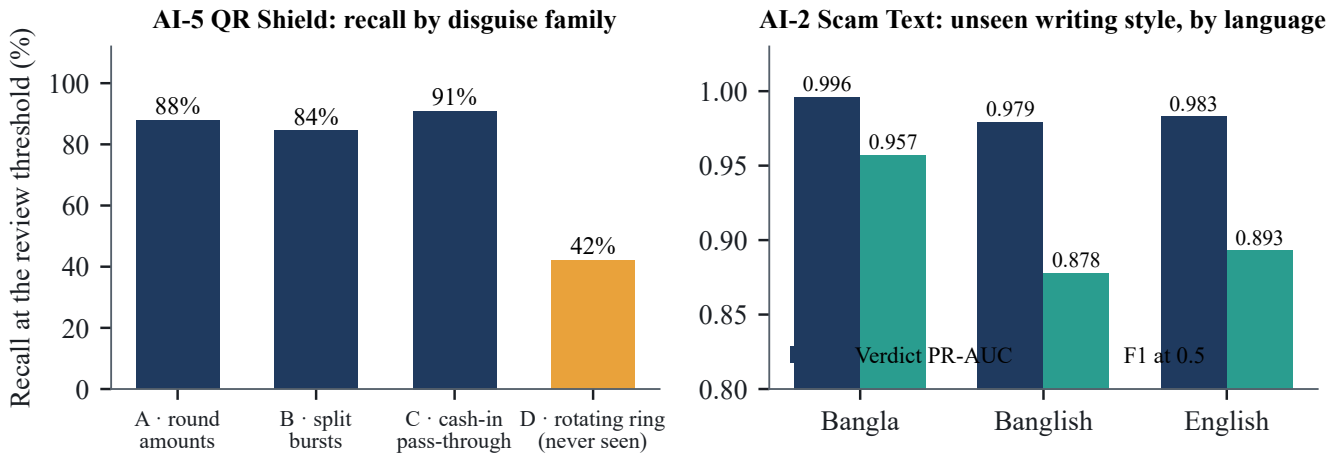


Figure 8. Left: AI-5 recall by disguise family; family D was hidden from training. Right: AI-2 by language on the unseen writing style (served model).

8.5. AI-6 Case Linker and AI-7 Grounded Brief

AI-6 joins **377** alerts (151 of them real scams) into **197** cases, 47.7% fewer items to open. Every pair of real scam alerts paid to the same mule was linked (365 of 365 pairs), mean case purity is 0.90, and 72.4% of the mules’ cash-outs are found inside the detected cases. Ranked by chain score, the top 5 cases hold **94.0%** of the real scam alerts (top 10: 96.0%; top 20: 98.0%). The top case is a mule ring with 130 alerts, 109 victims, 34 mule wallets, 28 shops and 39 cash-out points (Fig. 9e).

For AI-7, Gemini’s output passed the validator for **97.9%** of 140 briefs; the other 2.1% were replaced by templates, so 100% of the text shown to users was valid. In 30 prompt-injection attacks hidden in customer notes, the raw Gemini output was unsafe in 10% of cases, and **none** reached a user (Table 8). During the run two Gemini models ran out of free quota; they were paused automatically and the remaining 194 calls completed without failure.

Table 8. AI-6 linking and AI-7 grounding metrics (NB06, NB07).

AI-6 Case Linker		AI-7 Grounded Brief	
Alerts → cases	377 → 197	Gemini output passing the validator	97.9%
Same-mule alert pairs linked	100% (365)	Template fallback	2.1%
Mean case purity	0.90	Shown text valid	100%
Cash-outs found for detected cases	72.4%	Injection: raw output unsafe	10.0%
Real scam alerts in top 5 / 10 / 20 cases	94 / 96 / 98%	Injection: shown to a user	0.0%
Louvain communities (largest)	182 (52 nodes)	Gemini calls / failures	194 / 0

8.6. Fairness

In the spirit of equality of opportunity [39], Table 9 reports AI-1 error rates by group on the test window at the pause threshold. False-positive rates stay between 0.4% and 1.6% in every group; customers with less than 90 days of tenure see the most false pauses (1.6%), as tenure is a risk signal. Missed-scam rates vary more (10–31%), highest for remittance receivers and long-tenure customers; we report these gaps as monitoring items for a pilot rather than hiding them. Calibration error is at most 0.003 in every group. No sensitive attribute is used or invented.

Table 9. AI-1 fairness slices (test window, pause threshold). FPR: false-positive rate; FNR: missed scams.

Slice	Group	n	FPR	FNR	Slice	Group	n	FPR	FNR
Zone	rural	20,954	0.9%	18.0%	Tenure	under 90 days	6,713	1.6%	18.6%
Zone	semi-urban	25,293	0.8%	24.6%	Tenure	90–365 days	23,651	0.9%	19.2%
Zone	urban	37,308	0.8%	26.4%	Tenure	over 365 days	53,191	0.7%	29.0%
Channel	app	58,411	0.8%	25.3%	Segment	daily wage	24,656	1.0%	21.8%
Channel	USSD	25,144	1.0%	20.1%	Segment	remittance	14,984	0.7%	31.2%
Language	Bangla	33,100	0.9%	21.5%	Segment	salaried	36,639	0.9%	27.3%
Language	Banglish	29,098	0.8%	22.2%	Segment	student	7,276	0.4%	10.1%
Language	English	21,357	0.8%	28.1%					

9. The Implemented Prototype: Key Features

The prototype is a working website at <https://uvera-ai.vercel.app> backed by the FastAPI service (Fig. 9). Table 10 lists the key features by user and the AI behind each one.

Table 10. Key features of the prototype.

User	Feature	What the person sees	AI
Customer	Pause Check	A risk dial, three plain reasons, four choices (wait 10 minutes, verify, ask a trusted person, send anyway), Bangla or English, “not sure” when unsure	AI-1, AI-7
Customer	Scam Text Check	Scam probability, the scam family, highlighted phrases, a warning when a message tries to instruct an AI	AI-2
Customer	Cash-Flow Guardian	7-day forecast band, chance of running low, three savings plans; nothing moves automatically	AI-3
Agent	Cash to hold	Cash for a 90%-safe day, extra above usual, peer comparison, area-level QR pressure (never shop names)	AI-4, AI-5
Analyst	Case queue and workspace	Cases ranked by risk and deadline, replayable money-path graph, evidence E1–E5, Gemini brief, Dispute Clock, decision with a written reason	AI-6, AI-7
Analyst	QR watchlist	Shops in review, watch, not-sure and normal states with their top reason	AI-5
Judges	Trust Center	Every metric with its notebook source, fairness slices, model choice, limitations	all

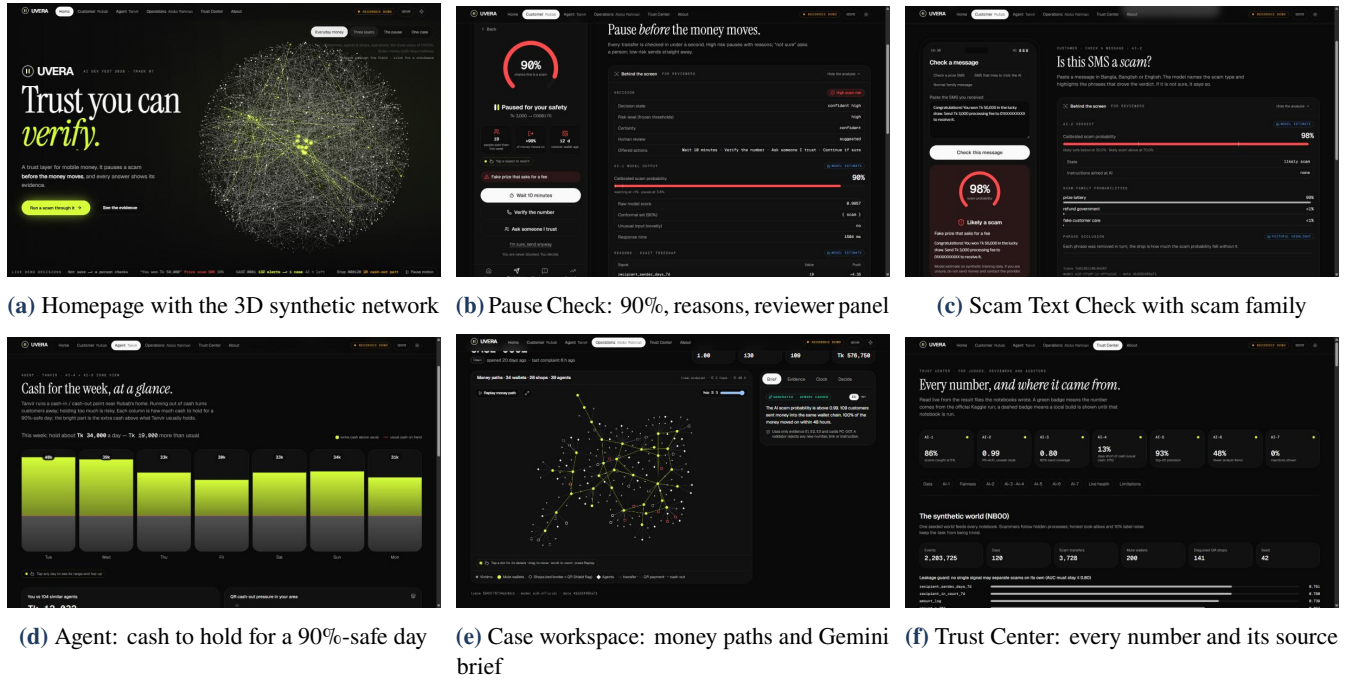


Figure 9. Screens of the working prototype (<https://uvera-ai.vercel.app>). All people, wallets and shops are synthetic.

10. Responsible AI, Privacy and Security

Table 11 maps each minimum expectation of the guideline’s responsible-AI requirements to UVERA’s design and its evidence.

Table 11. Responsible-AI requirements and how UVERA meets them.

Principle	UVERA design	Evidence
Privacy	Only synthetic data and one open dataset; no real customer record; only synthetic text is sent to Gemini	Data card; 0 real records
Explainability	Exact TreeSHAP reasons, phrase occlusion, evidence graphs, cited briefs	Reasons on every screen
Fairness	Error rates by zone, channel, tenure, language and segment; QR false positives by shop size	Table 9; peer-size comparison
Security	Prompt-injection detection and validator, JWT access control, per-visitor rate limits, no secrets in code, leakage guard	0% injections shown; threat model
Human oversight	Never blocks; “not sure” goes to a person; holds and case decisions need a person with a written reason	Policy rules; audit log
Transparency	Every output labelled as model estimate, business rule or generated wording; model cards [40], data card [41]; Trust Center	Labels in the interface
No harmful automation	No automatic denial, merchant closure or credit decision	Decision policy

11. Real-Life Impact and Business Value

Customers. At the same alert budget, UVERA finds three times as many scams as a common rule with fewer false pauses. In the synthetic test window, the scam money it flags is Tk 692,660 against Tk 437,300 for the rule; if 30–70% of warned customers stop (an assumption), the money kept by customers would be Tk 207,798–Tk 484,862 against Tk 131,190–Tk 306,110 for the rule. The pause costs a normal customer almost nothing: 99.1% of normal transfers pass without a pause, and nobody is ever blocked.

Agents. Holding the forecast amount reduces days without enough cash from 21.0% to 12.6%, so fewer customers are turned away. We also report that spreading the same extra cash evenly performs almost as well (12.9%); the

forecast’s value is in deciding *how much* extra cash to hold, which is the decision agents actually make.

Operations. Analysts open 47.7% fewer items, and the first five cases already contain 94% of the real scam alerts, each with a money-path graph and a deadline clock aligned with the 2026 Bangla QR dispute rules [4, 5].

Ecosystem. QR Shield finds shops that turn QR payments into cash to avoid cash-out charges (publicly listed at about Tk 13–18.50 per Tk 1,000, that is 1.30–1.85%; UVERA assumes 1.5% [42]); 92.5% of its weekly review list are real cases, and agents only see area-level totals.

Economics. The served models run on a CPU-only computer (the API peaks at about 1.2 GB of memory) with no GPU; only the optional Gemini brief calls an external service (median 6.5 s, with an instant template fallback). The prototype runs at near-zero cost on free tiers, which matters for an inclusive service with thin margins.

12. Scalability, Integration and Path to Real Data

Integration. UVERA plugs into a wallet through a small API: the app sends a draft transfer and receives a decision, reasons and a brief before showing the Confirm screen; the same API serves the agent and operations tools. Business rules change in configuration files without retraining, and models are exported as portable text and JSON, not pickles.

Path to real data. Table 12 follows the guideline’s post-hackathon pathway. Every number in this report would be re-measured on governed real data before any claim about real-world accuracy is made.

Table 12. Post-hackathon pathway for UVERA.

Stage	UVERA plan
Technical review	Code, notebooks, threat model and model cards are public; every run is reproducible from a commit.
Business review	Measure money kept by customers, false-pause cost, agent stock-outs and analyst hours per case.
Controlled validation	Retrain the same pipeline on governed, de-identified data; keep time- and customer-disjoint testing.
Proof of concept	Shadow mode: score real transfers without showing pauses; compare with confirmed outcomes.
Pilot	Show pauses to a small, consenting group; monitor fairness slices and recalibrate thresholds monthly.
Decision	Integrate, extend, or stop, based on measured scams stopped and false pauses.

13. Limitations

- **Synthetic data.** The scam families and mule behaviour were designed by the team; real scams may be harder. Label noise and honest look-alikes reduce, but do not remove, this risk.
- **New disguise styles.** QR Shield catches only 42.1% of the never-seen rotating ring; new patterns need analyst feedback.
- **Real-world text.** The served text model is weak on real English advertising spam (UCI PR-AUC 0.224); real messages would be needed before deployment.
- **Forecast limits.** Day-level stock-outs are mostly chance in our world, so the product plans cash rather than flagging days.
- **Language model.** Gemini’s free tier has quotas and its raw output can be manipulated; UVERA relies on the validator and templates, and never lets the model decide.

14. Conclusion

UVERA shows that a small, honest set of AI components can protect people at the moment a scam happens without taking the decision away from them. It pauses risky transfers with calibrated probabilities and plain reasons, says “not sure” when it should, plans agent cash close to the best that is achievable, turns hundreds of alerts into a few evidence-backed cases, and keeps a language model inside strict limits. Every result is measured against a simpler baseline on data the models never saw, and every number can be traced to a notebook run. The next step is to repeat the same measurements on governed real data in a controlled pilot.

Team Contributions

Member	Contribution
Tanvir Hossin	Team lead; product story, user research framing, visual and interaction design, video and presentation
Abdullah Rubab	Lead AI/ML and development: synthetic world, AI-1 to AI-7, evaluation, Kaggle notebooks, website and API
Abdur Rahman	Backend services and AI/ML: API endpoints, model notebooks, testing and deployment support

Data and Code Availability

All code, notebooks, model cards, the data card and the threat model are available at <https://github.com/ABRUBAB/takaresolve-ai-hackathon-2026> under the MIT licence. The synthetic world is regenerated from seed 42 by notebook NB00 and verified by hash; the official result files are in `reports/`, and every model file is listed with its SHA-256 in `artifacts/manifest.json`. The live prototype is at <https://uvera-ai.vercel.app>.

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